$\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y} \qquad \boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$

- Real eigenpairs $(\lambda_i, \boldsymbol{v}_i)$ give the real solution

$$\boldsymbol{y}_i(t) = e^{\lambda_i t}\boldsymbol{v}_i.$$

- Complex conjugate pairs $\lambda_\pm = \alpha \pm \beta i$ and $\boldsymbol{v}_\pm = \boldsymbol{a} \pm \boldsymbol{b}i$ give the two real solutions

$$\begin{aligned} \mathrm{re}\left(e^{\lambda_+ t}\boldsymbol{v}_+\right) &= e^{\alpha t}(\boldsymbol{a}\cos(\beta t) - \boldsymbol{b}\sin(\beta t)), \\ \mathrm{im}\left(e^{\lambda_+ t}\boldsymbol{v}_+\right) &= e^{\alpha t}(\boldsymbol{a}\sin(\beta t) + \boldsymbol{b}\cos(\beta t)). \end{aligned}$$

- The real general solution is $\boldsymbol{y}(t) = c_1\boldsymbol{y}_1(t) + \cdots + c_n\boldsymbol{y}_n(t)$
- What can go wrong?

Missing Eigenvectors $y' = A_\varepsilon y, \quad A_\varepsilon = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda+\varepsilon \end{pmatrix}$

The eigenvalues and eigenvectors are

$$\lambda_1 = \lambda, \qquad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$
$$\lambda_2 = \lambda + \varepsilon, \quad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix}.$$

Note $\lambda_2 \to \lambda$ and $\boldsymbol{v}_2 \to \boldsymbol{v}_1$ as $\varepsilon \to 0$. The solutions

$$\boldsymbol{y}_{1,\varepsilon}(t) = e^{\lambda t}\boldsymbol{v}_1 \qquad \xrightarrow{\varepsilon\to 0} e^{\lambda t}\boldsymbol{v}_1,$$
$$\boldsymbol{y}_{2,\varepsilon}(t) = e^{(\lambda+\varepsilon)t}\boldsymbol{v}_2 \quad \xrightarrow{\varepsilon\to 0} e^{\lambda t}\boldsymbol{v}_1.$$

We need two linearly independent (LI) solutions. The limit solutions are **not** LI.

A Second Solution

$$\frac{\boldsymbol{y}_{2,\varepsilon}(t)-\boldsymbol{y}_{1,\varepsilon}(t)}{\varepsilon}=e^{\lambda t}\begin{pmatrix}(e^{\varepsilon t}-1)/\varepsilon\\ e^{\varepsilon t}\end{pmatrix}\xrightarrow{\varepsilon\to 0}e^{\lambda t}\begin{pmatrix}t\\1\end{pmatrix}$$

- The title scaled difference solves $\boldsymbol{y}'=\boldsymbol{A}_\varepsilon\boldsymbol{y}$. The limit is from L'Hôpital's rule.
- $\boldsymbol{y}_2(t)=e^{\lambda t}(t\boldsymbol{v}_1+\boldsymbol{w})=e^{\lambda t}\left(t\begin{pmatrix}1\\0\end{pmatrix}+\begin{pmatrix}0\\1\end{pmatrix}\right)$ satisfies $\boldsymbol{y}'=\boldsymbol{A}_0\boldsymbol{y}$.
- To make this construction systematic substitute $\boldsymbol{y}_2=e^{\lambda t}(t\boldsymbol{v}_1+\boldsymbol{w})$ into $\boldsymbol{y}'=\boldsymbol{A}\boldsymbol{y}$

$$\boldsymbol{y}_2'=\lambda e^{\lambda t}(t\boldsymbol{v}_1+\boldsymbol{w})+e^{\lambda t}\boldsymbol{v}_1=e^{\lambda t}(\lambda t\boldsymbol{v}_1+\lambda\boldsymbol{w}+\boldsymbol{v}_1)$$

$$\boldsymbol{A}\boldsymbol{y}_2=e^{\lambda t}(t\boldsymbol{A}\boldsymbol{v}_1+\boldsymbol{A}\boldsymbol{w})=e^{\lambda t}(\lambda t\boldsymbol{v}_1+\boldsymbol{A}\boldsymbol{w})\quad\text{since }\boldsymbol{A}\boldsymbol{v}_1=\lambda\boldsymbol{v}_1$$

$$\boxed{\boldsymbol{y}_2'=\boldsymbol{A}\boldsymbol{y}_2}\implies\boxed{\lambda t\boldsymbol{v}_1+\lambda\boldsymbol{w}+\boldsymbol{v}_1=\lambda t\boldsymbol{v}_1+\boldsymbol{A}\boldsymbol{w}}$$

$$\implies\boxed{\lambda\boldsymbol{w}+\boldsymbol{v}_1=\boldsymbol{A}\boldsymbol{w}}\implies\boxed{(\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{w}=\boldsymbol{v}_1}$$

- Row reduce $\left[\begin{array}{c|c}\boldsymbol{A}_0-\lambda\boldsymbol{I}&\boldsymbol{v}_1\end{array}\right]=\left[\begin{array}{cc|c}\begin{pmatrix}\lambda&1\\0&\lambda\end{pmatrix}-\lambda\begin{pmatrix}1&0\\0&1\end{pmatrix}&\begin{pmatrix}1\\0\end{pmatrix}\end{array}\right]=\left[\begin{array}{cc|c}0&1&1\\0&0&0\end{array}\right]$.

  So $w_2=1$ and we can choose $w_1=0$ to get the title solution.

Missing Eigenvector for $y' = Ay$: $Av_1 = \lambda_1 v_1,\ v_1 \neq 0$ λ_1 repeated

Substitute $y_2(t) = e^{\lambda_1 t}(tv_1 + w)$ if missing an eigenvector for a double eigenvalue λ_1.

$$
\begin{aligned}
y_2' &= \lambda_1 e^{\lambda_1 t}(tv_1 + w) + e^{\lambda_1 t} v_1 \\
&= e^{\lambda_1 t} \boxed{\lambda_1 t v_1 + \lambda_1 w + v_1} \\
Ay_2 &= e^{\lambda_1 t}(tAv_1 + Aw) \\
&= e^{\lambda_1 t} \boxed{\lambda_1 t v_1 + Aw} \quad \text{since } Av_1 = \lambda_1 v_1 \\
\boxed{y_2' = Ay_2} &\implies \boxed{\lambda_1 t v_1 + \lambda_1 w + v_1 = \lambda_1 t v_1 + Aw} \\
&\implies \boxed{\lambda_1 w + v_1 = Aw} \implies \boxed{(A - \lambda_1 I) w = v_1}
\end{aligned}
$$

Row reducing $\left[\begin{array}{c|c} A - \lambda_1 I & v_1 \end{array}\right]$ gives w. The 2nd solution is $\boxed{y_2(t) = e^{\lambda_1 t}(tv_1 + w)}$.

Double Eigenvalues: Matrices to Row Reduce

Row reduce to find $\boldsymbol{v}_1$:

$$\left[\begin{array}{c|c} \boldsymbol{A} - \lambda_1 \boldsymbol{I} & \boldsymbol{0} \end{array}\right], \qquad (\boldsymbol{A} - \lambda_1 \boldsymbol{I})\boldsymbol{v}_1 = \boldsymbol{0}.$$

Row reduce to find $\boldsymbol{w}$:

$$\left[\begin{array}{c|c} \boldsymbol{A} - \lambda_1 \boldsymbol{I} & \boldsymbol{v}_1 \end{array}\right], \qquad (\boldsymbol{A} - \lambda_1 \boldsymbol{I})\boldsymbol{w} = \boldsymbol{v}_1.$$

Two linearly independent solutions are

$$\boldsymbol{y}_1(t) = e^{\lambda_1 t}\boldsymbol{v}_1, \qquad \boldsymbol{y}_2(t) = e^{\lambda_1 t}(t\boldsymbol{v}_1 + \boldsymbol{w}).$$

Worked Example $y' = Ay, \quad A = \begin{pmatrix} 4 & 1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = (\lambda + 2)(\lambda - 3)^2 = 0 \implies \boxed{\lambda_1 = 3,\ \lambda_2 = 3},\ \lambda_3 = -2$$

For $\lambda_3 = -2$, row reduce to find $\boldsymbol{v}_3$:

$$\left[\ \boldsymbol{A} - (-2)\boldsymbol{I}\ \middle|\ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}\ \right] = \left[\begin{array}{ccc|c} 6 & 1 & 0 & 0 \\ -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \implies \boldsymbol{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

So

$$\boxed{\boldsymbol{y}_3(t) = e^{\lambda_3 t}\boldsymbol{v}_3 = e^{-2t}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}$$

Worked Example: $\lambda_1 = 3 \quad \boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}, \quad \boldsymbol{A} = \begin{pmatrix} 4 & 1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

For $\lambda_1 = 3$, row reduce to find $\boldsymbol{v}_1$

$$\left[\begin{array}{c|c} \boldsymbol{A} - (3)\boldsymbol{I} & \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ 0 & 0 & -5 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \implies \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

There is one independent eigenvector and one solution

$$\boxed{\boldsymbol{y}_1(t) = e^{\lambda_1 t}\boldsymbol{v}_1 = e^{3t}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}$$

To find the second solution $\boldsymbol{y}_2(t) = e^{\lambda_1 t}(t\boldsymbol{v}_1 + \boldsymbol{w})$ row reduce to find $\boldsymbol{w}$

$$[\ \boldsymbol{A} - (3)\boldsymbol{I} \mid \boldsymbol{v}_1\ ] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ -1 & -1 & 0 & -1 \\ 0 & 0 & -5 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \implies \boldsymbol{w} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Worked Example: $\lambda_1 = 3 \quad \boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}, \quad \boldsymbol{A} = \begin{pmatrix} 4 & 1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

To find the second solution $\boldsymbol{y}_2(t) = e^{\lambda_1 t}(t\boldsymbol{v}_1 + \boldsymbol{w})$ row reduce to find $\boldsymbol{w}$

$$[\ \boldsymbol{A} - (3)\boldsymbol{I} \mid \boldsymbol{v}_1\ ] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ -1 & -1 & 0 & -1 \\ 0 & 0 & -5 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \implies \boldsymbol{w} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

We need $w_1 + w_2 = 1,\ w_3 = 0$. With $w_2 = 0$ the general solution is

$$\begin{aligned} \boldsymbol{y}(t) &= c_1 e^{3t}\boldsymbol{v}_1 + c_2 e^{3t}(t\boldsymbol{v}_1 + \boldsymbol{w}) + c_3 e^{-2t}\boldsymbol{v}_3 \\ &= c_1 e^{3t}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + c_2 e^{3t}\begin{pmatrix} t+1 \\ -t \\ 0 \end{pmatrix} + c_3 e^{-2t}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ &= \underbrace{\begin{pmatrix} e^{3t} & (t+1)e^{3t} & 0 \\ -e^{3t} & -te^{3t} & 0 \\ 0 & 0 & e^{-2t} \end{pmatrix}}_{\text{fundamental matrix } \Phi(t)} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \end{aligned}$$